Before each lecture, study the YouTube videos with the numbers stated in the right column of the above table.

Date	Topic	Study these videos before the lecture
Nov 2	1. **Introduction**	-
Nov 9	2. **Fundamentals of prob. theory and stat.**	1.2-1.5
Nov 16	3. **RNG, sampling procedures, simple sampling**	2.1-2.6
Nov 23	4. **General variance reduction techniques**	3.1-3.5
Dec 7	5. **Analog Monte Carlo criticality simulations**	4.1-4.7
Dec 14	6. **Introduction to SERPENT and XS Libraries**	5.1-5.9 + 5.X
Jan 18	7. **Convergence of the Fission Source**	6.1-6.3
Jan 25	8. **Monte Carlo Burnup Calculations**	7.1-7.5
Feb 1	9. **Geometry and Model Representation, Tracking**	8.1-8.4
Feb 8	10. **Non-Analog Monte Carlo Simulations**	9.1-9.5
Feb 15	11. **Variance Red. Techniques and Parallel Sim.**	X.1-X.4
Feb 29	**Group Project Presentations**	

Topic
1.1 What is Monte Carlo
1.2 Random variables
1.3 Probability density function
1.4 Cumulative distribution function
1.5 Elements of statistics
2.1 Random number generators
2.2 Inverse transform
2.3 Acceptance rejection
2.4 Simple sampling
2.5 Variance of the mean value
2.6 Confidence intervals
3.1 Figure of merit
3.2 Control variate
3.3 Correlated sampling
3.4 Stratified sampling
3.5 Importance sampling
4.1 Selection of random directions
4.2 Sampling the source

Topic
4.3 Transition kernel
4.4 Selection of interaction type
4.5 Selection of energy and direction
4.6 Fixed source simulations
4.7 Criticality simulations
5.1 Serpent general info
5.2 Serpent library
5.3 Serpent input intro
5.4 Serpent input surfaces
5.5 Serpent input materials
5.6 Serpent input cells
5.7 Serpent input lattices
5.8 Serpent input options
5.9 Serpent execution
5.X Serpent demo
6.1 Convergence of fission source
6.2 Bias in fission source
6.3 Optimisation of source convergence
7.1 Burnup simulations
7.2 Fuel cycle simulations
7.3 Explicit Euler burnup scheme
7.4 Predictor-corrector burnup scheme
7.5 Implicit Euler burnup scheme
8.1 Geometry representation
8.2 Conventional tracking in space
8.3 Virtual collisions
8.4 Delta tracking
9.1 Neutron statistical weight
9.2 Russian roulette rule
9.3 Implicit capture
9.4 Implicit fission
9.5 Tallying procedures
X.1 Exponential transformation
X.2 Geometry splitting
X.3 Weight window technique
X.4 Parallel simulations